

ChatGPT Is Actually My Friend: Understanding the Information Behavior of University Students Interacting with AI Chatbots

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Abstract

The rapid adoption of AI chatbots has sparked widespread debates about how their prevalence will impact higher education. This late-breaking work presents findings from an exploratory study comprised of a survey of 199 university students and 8 semi-structured follow-up interviews on university students' use and assessment of AI chatbots as an information resource compared with search engines for a variety of topics. We found university students from varied disciplines view chatbots as a valuable all-purpose information tool for most topics for several reasons, including their perceived versatility and single-answer response format and their provision of emotional support. We discuss the implications of these findings for design and future research directions.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**.

Keywords

chatbots; human-ai interaction; credibility assessment

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1 Introduction

Since the release of ChatGPT for public use in November 2022, there has been a surge of discourse about the implications of AI chatbots for university students. Some contend that chatbots benefit post-secondary students by providing immediate feedback [33], personalized learning plans [19], improved study skills and time management [15], and increased motivation [1]. Although AI chatbots have shown remarkable results in their ability to answer questions and conduct other language-related tasks with high accuracy,

there have also been reports of chatbots delivering wrong or confusing information [6] and performing with mixed reliability in different domains such as law [13], medicine [8, 10, 43], science [39], and financial reports [46]. To inform the design of systems, policies, and educational interventions that encourage productive use of AI tools in higher education, it is important to understand how university students think about these tools—what kinds of information they think chatbots are a good resource for and how they use them in their daily lives both in and outside of the classroom. This late-breaking work presents findings from an exploratory study comprised of a survey of 199 university students and 8 semi-structured follow-up interviews on their use and assessment of AI chatbots as an information resource for 13 topics.

2 Related Work

2.1 Novel Information Behavior and University Students

Whenever novel tools for information retrieval, access, and production become widespread, researchers hasten to understand emergent information behaviors. For example, HCI researchers have investigated information behaviors with respect to use and design of search engines [38], wikis [17], and social media [25]. University students are often a population of interest in research related to novel tool use as they tend to be both early adopters of and reliant on new technologies for information [18, 22, 37, 45]. Work at CSCW has investigated how chatbots can enhance computer science college students' study habits [42], and research on the benefits of chatbots to college students for second language practice [24], debugging code [44], and assignment assistance [41] has also shown promise. However, concerns related to the proliferation of plagiarism [28], cheating, and ethical questions [29] have also been raised. Building upon work that examines how students utilize chatbots in academic contexts [35], this study focuses on how university students use AI chatbots in their day-to-day lives.

2.2 Deciding If AI Tools Are a Good Information Resource

Information assessment is an enduring topic for HCI researchers and practitioners. In order to create systems that provide good cues for people to assess information, we must first understand how people judge information and how system characteristics influence these judgments. Recent work has shown post-secondary students report mixed levels of trust in chatbots when assisting with coursework [3, 12]. Additionally, work has shown that while students recognize the potential for the misuse of [2] and overreliance on

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chatbots [11], they also value their accessibility and time-saving strengths [2]. Although work has looked at student behavior with generative AI in specific coursework contexts, such as computer science [2, 23, 36], and recent work has explored chatbot features that may elicit student trust [4] researchers are still exploring what types of information students believe chatbots are well-suited to produce, as well as the conditions under which they trust chatbot-produced information.

2.3 How People Think About AI Tools

It is widely a accepted concept that when people use technology, their behavior is guided by a mental model of how it works. How people understand AI, then, is particularly important when humans rely on AI tools to perform important tasks [5]. In the case of information seeking, understanding how information was produced and is maintained are critical aspects of making a decision about whether to trust it [16], and generative AI is a very new way of producing information—some may even argue that what it produces is not information at all. Because generative AI tools are newly popular, little is known about common models of how AI creates text or where AI-generated information comes from. In this study, we aim to understand three interconnected aspects of university students' information behaviors with respect to AI chatbots:

- **RQ1** What kinds of information do university students seek using AI chatbots?
- **RQ2** How reliable do university students think AI chatbots are as a resource for different types of information?
- **RQ3** How do university students think AI chatbots work?

3 Method

We conducted an IRB-approved study that included a survey and semi-structured interviews. We first conducted an online survey through Qualtrics that asked university students about their information-seeking preferences for a variety of topics using chatbots, and as a comparison point, search engines (See [7] for additional discussion of students' use of chatbots as search). The topic list was adapted from a previous study on search behavior [17]. Survey participants had to be university students, at least 18 years old, and used an AI chatbot before. The first author recruited participants via social media. After survey completion, participants could enter a drawing for one of three \$100 Amazon gift cards. There were 799 survey responses in total completed in August 2023. We took multiple steps to ensure high data quality. We included a commitment check asking participants to commit to providing thoughtful, truthful answers [20]. We discarded surveys from participants who did not pass the commitment check, took under two minutes to complete, were incomplete, had nonsensical or repeated responses, or had duplicate usernames, resulting in a total of 199. We used R to perform chi-square tests to compare frequencies between responses for search engine and chatbot behavior and Fisher's pairwise comparisons with a Bonferroni correction for differences between majors to account for the smaller sample sizes. We analyzed open-ended question responses using a thematic inductive approach[9].

We recruited interview participants from the pool of students who agreed to be contacted for follow-up interviews in an end-of-survey question. Interviews were semi-structured with a think-aloud protocol [14]. We asked participants about their experiences searching for information with chatbots and to elaborate on their survey responses on searching for topics using a search engine and chatbot. Next, the interviewer shared her screen with a view of ChatGPT (v. 3.5) and asked the participant to verbalize a query about one of the survey topics that they would be likely to ask a chatbot, which the interviewer then typed into ChatGPT. The participant was then asked to explain their evaluation of the chatbot's response. The first author conducted 8 interviews with participants (4 undergraduate, 4 master's; 2 female, 6 male; 7 Black/African American, 1 White) over Zoom that were audio recorded and then transcribed by Rev.com. Interviews were 30-60 minutes long, and participants were compensated \$30 USD cash via Venmo or Amazon gift card. All interviews took place between October 2023 and March 2024. We analyzed the data using an inductive thematic analysis approach [9]. After familiarization with the transcripts, the first author and a member of the research team performed line-by-line iterative, inductive coding of all the interviews, producing a set of open codes, which were reviewed with the research team regularly. The group collaboratively constructed a shared interpretation of the results through discussion as well as iterative writing, rewriting, and restructuring of the results.

4 Findings

4.1 What kinds of information do university students seek using AI chatbots?

We surveyed participants on whether they had searched for, would search for (but had not), or would not search for 13 information topics using a search engine and a chatbot. Overall, a high percentage of participants reported that they have or would search for all of the types of information they were asked about using either a search engine or a chatbot (see Table 2). Surprisingly, chi-square tests revealed no statistically significant differences between search engine and chatbot search behavior when comparing any of the information domains for the three choices *Have searched*, *Would search*, or *Would not search* ($p < 0.05$). We anticipated there would be significant differences among domains, which would provide grist for reflecting on the relative advantages or disadvantages of chatbots as an information resource. For example, we expected that participants might be less likely to use a chatbot for medical information or legal advice because they would prefer to find a trusted resource and the query might contain sensitive information. However, the results for these topics did not significantly differ for search engine and chatbot search behavior. Given the fairly high representation of computer and information science students in our participant pool, we also used Fisher's pairwise comparison tests to see if CS/IS student chatbot search behavior was different from the other majors; yet, we only found one—CS/IS students chose *Would search* significantly more than *Would not search* for Educational Opportunities using chatbots than non-CS/IS majors ($p = 0.031$).

Although survey data yielded no statistically significant differences between chatbot and search engine preferences, interview

Gender	Age	Degree Level	Major
Woman: 90 (43.14%)	18-21: 43 (22.06%)	Associates: 16 (7.84%)	Business/Economics: 11 (5.53%)
Man: 108 (54.90%)	22-25: 70 (34.80%)	Undergrad.: 130 (64.71%)	Computer/Info. Science: 86 (43.21%)
Non-binary: 0 (0%)	26-29: 45 (22.55%)	Master's: 37 (19.61%)	Engineering: 20 (10%)
Didn't disclose: 1 (0.49%)	30-49: 38 (19.12%)	PhD: 16 (7.84%)	Natural/Physical Sciences: 11 (5.53%)
Self-described: 0 (0%)	50-64: 2 (0.98%)		Soc. Sciences/Humanities: 50 (25.12%)
	65+: 1 (0.49%)		Other: 19 (9.55%)
			None: 2 (1.01%)

Table 1: Total number and percentage of respondents by gender, age, degree level, and major (N=199)

data provided insights about information topics about which students preferred a chatbot or search engine. Sometimes, participants found search engine features better suited to the task. For example, P7 and P8 preferred using a search engine for travel research because the chatbots they used lack images. Overall, as indicated in the survey results, interview participants did not differentiate between domains. When P6 was asked if there was any type of information that he wouldn't use a chatbot for, he disclosed, "I will say at any time I'll use [a chatbot], at any time, any topic."

4.2 How reliable do university students think AI chatbots are as an information resource for different types of information?

We asked how reliable survey participants believe chatbots are among those who *had searched* or *would search* for information within each of the listed domains. Overall, results were positive with every domain receiving a total of 50% or higher of *excellent* and *good* ratings, and less than 13% rated any domain *poor* or *very poor* (Figure 1). Building on the data from the survey responses, we used the interviews as an opportunity for participants to explain their willingness to use chatbots to search for various topics and to share their personal experiences searching for information with chatbots. Five of the participants (P1, P2, P3, P4, P6) shared a common reason for their confidence in chatbots: they provide one straightforward and precise response that they deemed to be more efficient than sifting through the clutter of potential answers that are returned by search engines. P2 explained that: "Chatbots will actually go straight to the point to give you what you are actually looking for and analyze it the way you actually understand it." Similarly, P4 stated, "It's like a search engine, but it's more precise and concise with what you want from it." And finally, P6 explained: "Yeah, like I say, ever since I was introduced to chatbots, I actually make use of more chatbots than search engines...because I can actually get some precise and cool answer."

4.3 How do university students think AI chatbots work?

We asked interview participants to explain how chatbots work. Participants often compared chatbots to humans when describing their abilities (P1, P2, P3, P4, P6). P4 compared chatbots to humans when explaining their versatility: "I think they're more like humans. They do a million things." Other times, participants used human comparisons to justify chatbots' limitations. For example, P6 explained why

ChatGPT can't answer questions requiring personal information: "So, it wasn't a very big surprise because, just like I said, a chatbot is actually more related to a human [and] I'm not sure that there is any human on Earth that can answer the question."

Finally, several of our participants disclosed how they value the emotional support chatbots provide along with informational support (P1, P4, P6). P4 explained, "The chatbot will actually give you the answer and you can also relate with the chatbot...let's say the environment was so quiet, and you need somebody to talk to." At times, the human-like conversational aspects of the chatbot responses appeared to signal to participants that the chatbot had positive intentions toward them. P6 shared, "[ChatGPT] is actually my friend here because I can actually log in and say, 'Good morning,' and it'll say, 'Morning, how are you, how may I be of help today?'"

5 Discussion: Omniscient Oracles: implications of using chatbots as all-purpose information resources

In our survey data, there was no statistical difference between university students' reported search behaviors and goals in using search engines and AI chatbots for any of the survey topics. In short, when it comes to chatbot utility and reliability as an information resource, they do not differentiate between topic areas. Furthermore, our data pointed to students' overall confidence in chatbots to deliver an objectively accurate response in *all* domains, suggesting a widespread belief in something like omniscience. These findings are noteworthy for a number of reasons.

First, the accuracy of information from chatbots is variable. A bevy of research looks at accuracy rates of chatbots in different domains and finds inconsistencies [21, 26, 39]. Moreover, AI chatbots may provide students with prejudiced responses as a consequence of bias in training data [31, 40]. The reliance on one type of tool that is prone to error and bias as an all-purpose resource in higher education clearly raises concerns. Yet, these may be the least worrisome aspects of our findings. The speed of advances in foundation models is staggering and research on the accuracy and bias of information they generate likely has a short shelf life.

Our data provided insights not only about what student use chatbots for, but about why they use them and how they made sense of them. Interview participants valued chatbots' single-answer format over the perceived jumble of search engine results, a feature that also surfaced in a recent Extended Abstract at CHI [27]. The absence of multiple results to a query signaled to our participants

Topic	Have searched		Would search		Would not search	
	SE	AI	SE	AI	SE	AI
Educational opportunities	56.28%	49.75%	32.2%	32.2%	11.56%	18.1%
Family relationships	50.25%	47.24%	36.2%	37.7%	8.04%	15.1%
Fitness	56.78%	56.78%	39.7%	35.2%	3.52%	8.0%
Home and garden	43.72%	42.21%	42.2%	44.7%	14.07%	13.1%
Job responsibilities	56.78%	46.73%	32.2%	40.2%	11.06%	13.1%
Legal advice	49.25%	45.73%	35.7%	42.2%	15.08%	12.1%
Medical treatments	58.79%	61.81%	34.2%	30.2%	7.04%	8.0%
Mental health	56.28%	52.26%	32.2%	38.2%	11.56%	9.5%
Personal finances	45.73%	45.23%	41.2%	40.2%	13.07%	14.6%
Politics	44.22%	37.19%	42.7%	46.2%	13.07%	16.6%
Product reviews	54.77%	50.75%	37.2%	36.2%	13.57%	13.1%
Sexual practices	40.70%	35.68%	40.7%	46.7%	18.59%	17.6%
Travel planning	54.77%	53.27%	31.7%	36.2%	13.57%	10.6%

Table 2: Percentage of survey participants' answers about searching for a topic with a search engine (SE) or an AI chatbot (AI) (N=199)

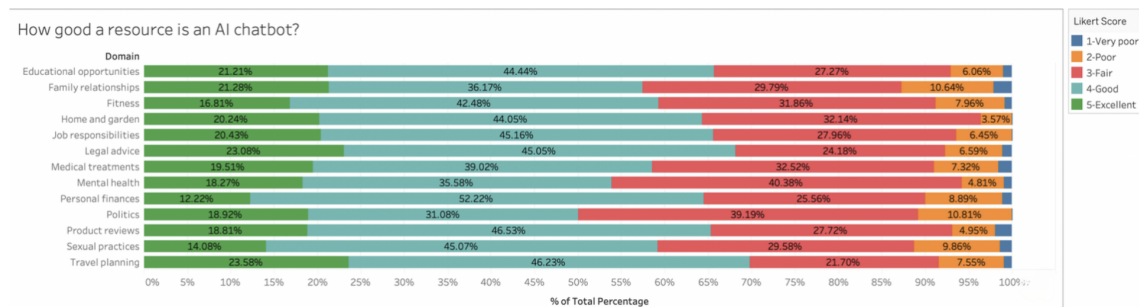


Figure 1: Distribution of Likert scores for "Please rate the extent to which you believe AI tools are a good resource for this information." Participants were only presented with this question for domains that they reported they had or would search for using chatbots.

that the single result they were seeing was the sole correct one. Search engine and chatbot interfaces offer experiences that manifest fundamentally different epistemological orientations. This raises a critical question for information behavior researchers and interface designers: during a time of heightened algorithmic curation that facilitates echo chambers, is one response the best delivery format, particularly for complex questions? University students are in an environment and phase of life in which they are called on to develop critical thinking and analytical skills, and the ability to evaluate different sources of information is key to growing this skill set. Exploring chatbot designs that offer multiple answers from diverse perspectives offers the possibility of intriguing experiments on stimulating critical thinking.

Finally, participants frequently anthropomorphized AI chatbots when praising their strengths or defending their limitations. Some interview participants shared that they interpreted AI chatbots' convivial virtual personalities as evidence that they had benevolent intentions, with one participant referring to ChatGPT as "his friend." If people confer human-like attributes to chatbots while using them as all-purpose information resources, this opens up a trove of potential theoretical perspectives and empirical questions

drawing on communication and psychology traditions. Our study focused on information seeking, yet some participants championed the human-like conversational abilities to provide emotional as well as informational support. Although some work has explored the use of chatbots by university students to bolster mental health and well-being [30, 32], the idea that an all-purpose information resource can create a sense of interpersonal connection and support not only suggests leveraging psychology and communication literature to understand credibility assessment practices, but also how social influence and reciprocity in information exchange may influence human-AI interaction. The addition of an emotional connection to an anthropomorphized agent with perceived omniscience and desire to help users suggests that new kinds of information behaviors are on the horizon.

6 Limitations and Acknowledgments

Neither our survey nor interview sample is representative of a defined university student population. In five interviews, we witnessed discrepancies between the race and age the participant reported in their survey and the self-reported demographics in the interviews. Due to a lack of confidence in the validity of the rest of

the data associated with the participants who reported conflicting demographic information, we dropped data from these participants from all analyses (See [34] for a broader discussion of fraudulent participation in online research). We also note that chatbot technology is rapidly advancing, and the chatbots and features present in our data are limited to those available to participants at the time. Finally, we would like to thank Anindita Bandyopadhyay for her contributions to this research.

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